

Beginning Algebra Practice Test I Version 2

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show work when necessary.
- iii) Draw a rectangle around your answers.
- iv) Staple a copy of these test questions with your work and your final answers.

Simplify the following polynomial, write in descending order of powers, and state its degree.

① $10x^3 + 5x^3 - 6 + 3x - 2x + 3x^2 - 8 + x^3 + 1$

② Add the following polynomials.

a) $(-4x^2 + 2) + (3x^3 - 4x^2 + 17)$

b) $(8x^2 - x - 10) + (-x^2 - 20x + 3)$

③ Subtract the following polynomials.

a) $(7x^2 + 4x + 9) - (-2x^3 - 2x^2 + x + 4)$

b) $(6x^4 - 2x^3 + x^2 - x + 3) - (3x^3 + 20x^2 + 4x)$

Multiply.

④ a) $4x^5(2x^3 + 8x^2 - 2x + 2)$ b) $-8x^2(-3x^3 - 2x + 1)$

Divide.

⑤ a) $\frac{33x^4 + 88x^3 + 55x^2}{11x^2}$ b) $\frac{14x^4 - 20x^3 + 8x^2}{-2x}$

Expand.

⑥ a) $(x+6)(x+3)$ b) $(2x-1)(7x+2)$

⑦ a) $(6a-7b)(2a-3b)$ b) $(3x-2)(4x^2+x-1)$

⑧ a) $(x-3)^2$ b) $(10a+b)^2$

Factor the following expressions completely.

⑨ a) $x^2-10x+16$ b) $9x^2+15x-14$

⑩ a) $x^2+12xy+20y^2$ b) x^2-16

⑪ a) a^2-49b^2 b) $-4+x^2$

⑫ a) $36-y^2$ b) $64-b^2$

⑬ a) $-x^2+1$ b) $81-4x^2$

⑭ a) x^4+x^2-30 b) $2x^4+9x^2-5$

⑮ a) $12x^4 + 5x^2 - 2$ b) $8x^3 - 32x^2 + 3x - 12$

⑯ a) $x^3 + 3x^2 - 25x - 75$ b) $b^5 - 2b^4 - b + 2$

⑰ $9x^3 + 18x^2 + x + 2$

⑱ $10x^3 - 1000x$

⑲ $-2x^4 + 30x^3 - 100x^2$

⑳ $80x^3 - 45x$

㉑ $x^5 - 37x^3 + 36x$

㉒ $36a^2b - 25b^3$

㉓ $24a^3b + 88a^2b^2 - 50ab^3$

$$\textcircled{24} 2y^3 + 3y^2 + 8y + 12$$

$$\textcircled{25} 80x^4y - 510x^3y^2 + 180x^2y^3$$

Answers for Version 2

① $16x^3 + 3x^2 + x - 13, D=3$

② a) $3x^3 - 8x^2 + 19$ b) $7x^2 - 21x - 7$

③ a) $2x^3 + 9x^2 + 3x + 5$ b) $6x^4 - 5x^3 - 19x^2 - 5x + 3$

④ a) $8x^8 + 32x^7 - 8x^6 + 8x^5$ b) $24x^5 + 16x^3 - 8x^2$

⑤ a) $3x^2 + 8x + 5$ b) $-7x^3 + 10x^2 - 4x$

⑥ a) $x^2 + 9x + 18$ b) $14x^2 - 3x - 2$

⑦ a) $12a^2 - 32ab + 21b^2$ b) $12x^3 - 5x^2 - 5x + 2$

⑧ a) $x^2 - 6x + 9$ b) $100a^2 + 20ab + b^2$

⑨ a) $(x-8)(x-2)$ b) $(3x+7)(3x-2)$

⑩ a) $(x+10y)(x+2y)$ b) $(x-4)(x+4)$

⑪ a) $(a-7b)(a+7b)$ b) $(x-2)(x+2)$

⑫ a) $(6-y)(6+y)$ b) $(8-b)(8+b)$

⑬ a) $(1-x)(1+x)$ b) $(9-2x)(9+2x)$

⑭ a) $(x^2+6)(x^2-5)$ b) $(2x^2-1)(x^2+5)$

⑮ a) $(3x^2+2)(2x-1)(2x+1)$ b) $(8x^2+3)(x-4)$

⑯ a) $(x-5)(x+5)(x+3)$ b) $(b-1)(b+1)(b^2+1)(b-2)$

⑰ $(9x^2+1)(x+2)$

⑱ $10x(x-10)(x+10)$

⑲ $-2x^2(x-5)(x-10)$

⑳ $5x(4x-3)(4x+3)$

㉑ $x(x-1)(x+1)(x-6)(x+6)$

㉒ $b(6a-5b)(6a+5b)$

23 $2ab(6a+25b)(2a-b)$

24 $(y^2+4)(2y+3)$

25 $10x^2y(8x-3y)(x-6y)$